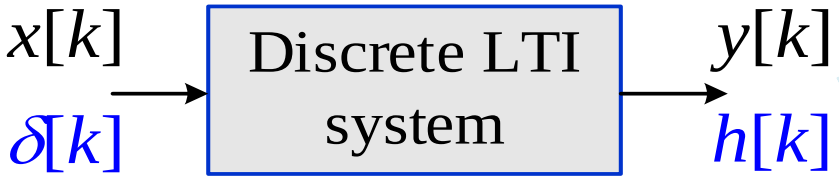
Digital Signal Processing

Zhou Hang

Frequency domain analysis of discrete periodic signals

- Why introduce frequency domain analysis of signals
- Representation of Discrete Periodic Signal DFS
- Properties of Discrete Periodic Signal DFS

Why introduce frequency domain analysis of signals



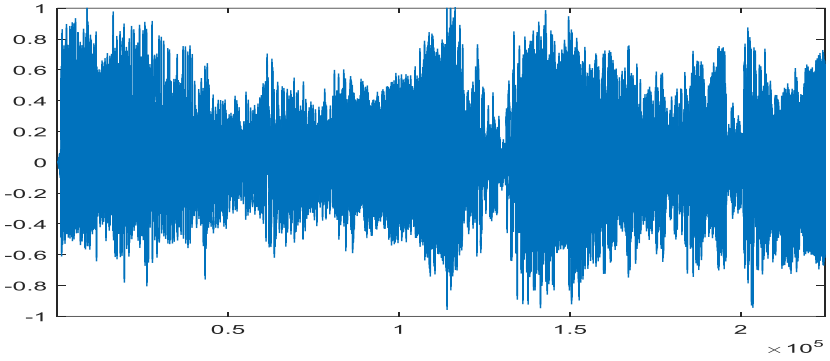
$$x[k] = \sum_{n=-\infty}^{\infty} x[n]\delta[k-n]$$

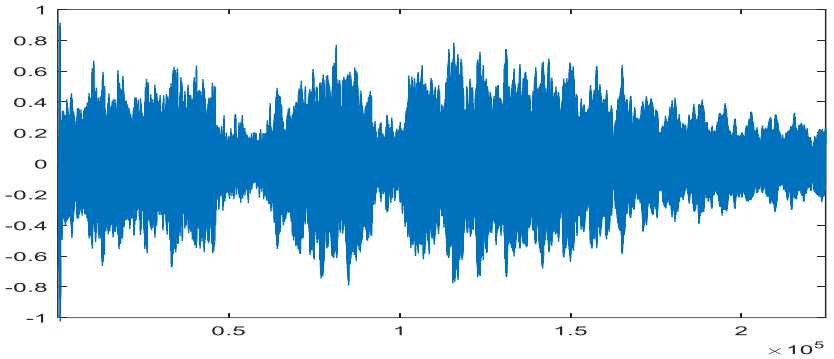
$$x[k] * h[k] = \sum_{n=-\infty}^{\infty} x[n]h[k-n] = y[k]$$

Why introduce frequency domain analysis of signals

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How to analyze the characteristics of sounds in different vocal ranges?

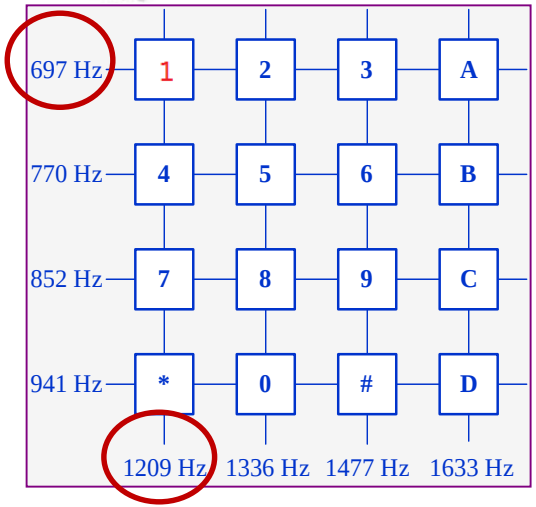




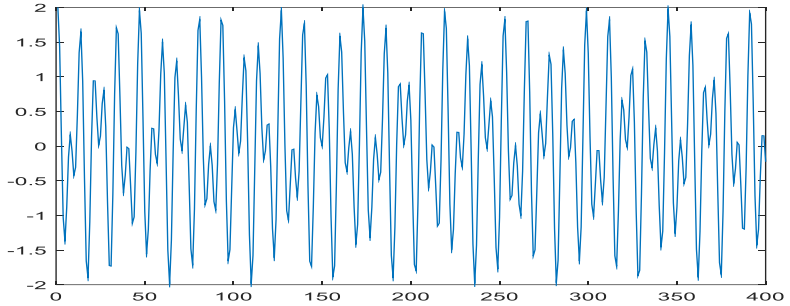
Why introduce frequency domain analysis of signals

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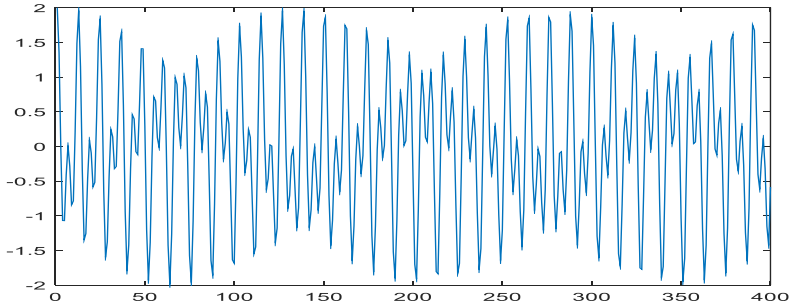






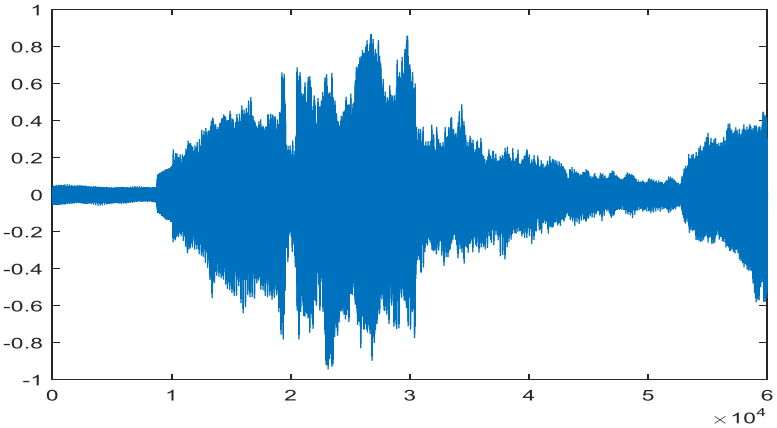




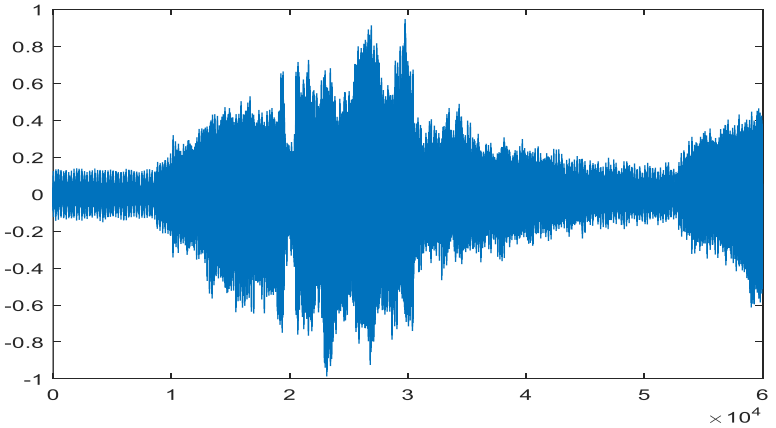


Why introduce frequency domain analysis of signals

How to filter out noise in the signal?



Time domain waveform of the signal

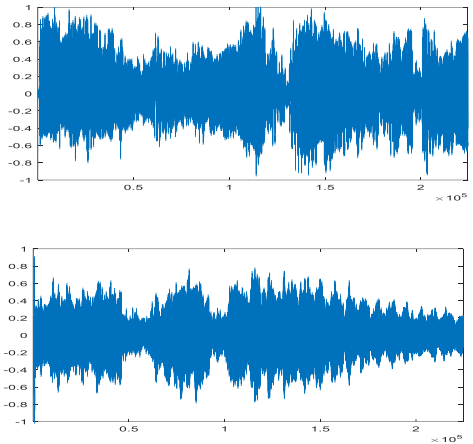


Time domain waveform of noisy signal

Why introduce frequency domain analysis of signals

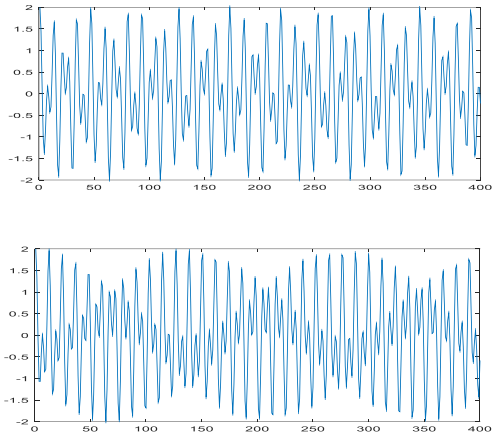
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Analyze sounds in different vocal ranges



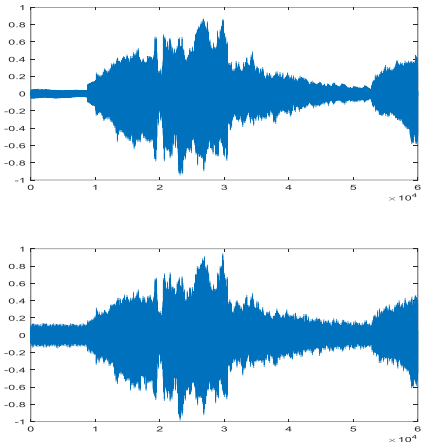
Time domain waveform of male bass/male treble

Identify telephone dialing keys



Time domain waveform of numeric keys' 1 'and' 2 '

Filter out noise in the signal

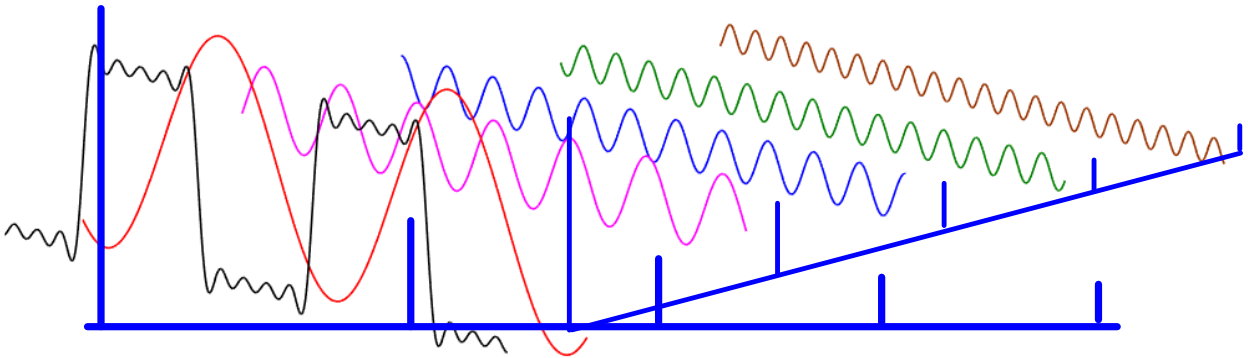


Time domain waveform of signal/noisy signal

There are limitations in signal time-domain analysis and processing

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n \, \mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$C_n = \frac{1}{T_0} \int_{t_0}^{T_0 + t_0} \tilde{x}(t) \mathrm{e}^{-\mathrm{j}n\,\omega_0 t} \mathrm{d}t$$



Any periodic sequence with a period of N, $\tilde{x}[k]$ can be represented by N imaginary index sequences:

$$\tilde{x}[k]=\frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}[m]\mathrm{e}^{\mathrm{j}\Omega_0 mk}$$

$$\Omega_0=\frac{2\pi}{N}$$

The weighting coefficients $\tilde{X}[m]$ in front of the complex exponential sequence vary with different time-domain signals.

$$\tilde{X}[m]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\Omega_0 mk}$$

DFS representation of discrete periodic signals

IDFS

$$\tilde{x}[k]=\frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}[m]\mathrm{e}^{\mathrm{j}\Omega_0 mk} \qquad k=0,1,\cdots,N-1$$

DFS

(Discrete Fourier Series)

$$\tilde{X}[m]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\Omega_0 mk} \qquad m=0,1,\cdots,N-1$$

The frequency spectrum $\tilde{X}[m]$ of a discrete periodic signal $\tilde{x}[k]$.

$$\tilde{x}[k]\xleftrightarrow{\text{DFS}}\tilde{X}[m]$$

DFS representation of discrete periodic signals

$$\tilde{X}[m]=\left|\tilde{X}[m]\right| \mathrm{e}^{\mathrm{j}\varphi[m]}$$

Amplitude spectrum

Phase Spectrum

(1) $\tilde{X}[m]$ is a discrete spectrum

(2) $\tilde{X}[m]$ is a periodic sequence with a period of N

$$\tilde{X}[m+N]=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}(m+N)k}=\sum_{k=0}^{N-1}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}=\tilde{X}[m]$$

[E.g] Find the frequency spectrum $\tilde{X}[m]$ of a sequence $\tilde{x}[k]=\{\cdots,\underset{\downarrow}{1},2,3,4,\cdots\}$ with a period of 4

Solution:

$$\tilde{X}[m]=\sum_{k=0}^{3}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}mk}=\sum_{k=0}^{3}\tilde{x}[k]W_4^{mk}$$

$$W_4=\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}}$$

$$\tilde{X}[0]=\tilde{x}[0]+\tilde{x}[1]+\tilde{x}[2]+\tilde{x}[3]=10$$

$$\tilde{X}[1]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot1\cdot1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot1\cdot2}+\tilde{x}[3]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot1\cdot3}=-2+2\mathrm{j}$$

$$\tilde{X}[2]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot2\cdot1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot2\cdot2}+\tilde{x}[3]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot2\cdot3}=-2$$

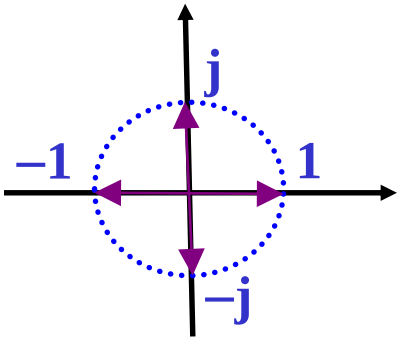
$$\tilde{X}[3]=\tilde{x}[0]+\tilde{x}[1]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot3\cdot1}+\tilde{x}[2]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot3\cdot2}+\tilde{x}[3]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{4}\cdot3\cdot3}=-2-2\mathrm{j}$$

[E.g] Find the frequency spectrum $\tilde{X}[m]$ of a sequence $\tilde{x}[k]=\{\cdots,\underset{\downarrow}{1},2,3,4,\cdots\}$ with a period of 4

Solution: $\tilde{X}[m]=\sum_{k=0}^{3}\tilde{x}[k]W_4^{mk}$ Expressed in matrix form, it can be obtained

$$\begin{bmatrix}\tilde{X}[0]\\ \tilde{X}[1]\\ \tilde{X}[2]\\ \tilde{X}[3]\end{bmatrix}=\begin{bmatrix}W_4^0 & W_4^0 & W_4^0 & W_4^0\\ W_4^0 & W_4^1 & W_4^2 & W_4^3\\ W_4^0 & W_4^2 & W_4^4 & W_4^6\\ W_4^0 & W_4^3 & W_4^6 & W_4^9\end{bmatrix}\begin{bmatrix}\tilde{x}[0]\\ \tilde{x}[1]\\ \tilde{x}[2]\\ \tilde{x}[3]\end{bmatrix}$$

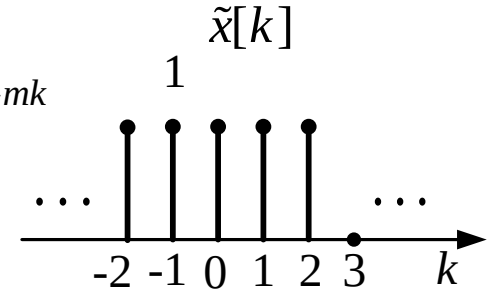
$$=\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & -j & -1 & j\\ 1 & -1 & 1 & -1\\ 1 & j & -1 & -j\end{bmatrix}\begin{bmatrix}1\\ 2\\ 3\\ 4\end{bmatrix}=\begin{bmatrix}10\\ -2+2j\\ -2\\ -2-2j\end{bmatrix}$$



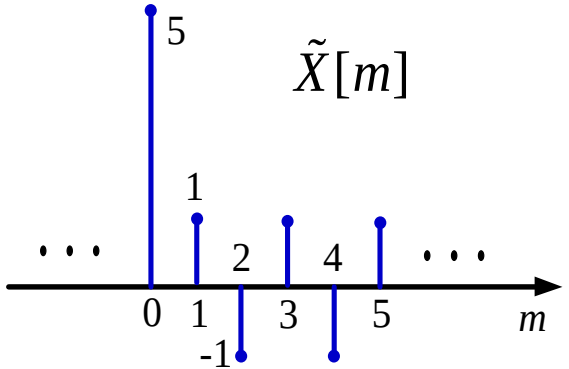
[E.g] Find the frequency spectrum of a periodic rectangular sequence with a period of N=6 as shown in the figure.

Solution:

$$\tilde{X}[m]=\sum_{k=-2}^{3}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\Omega_m k}=\sum_{k=-2}^{3}\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}=\sum_{k=-2}^{2}\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}$$

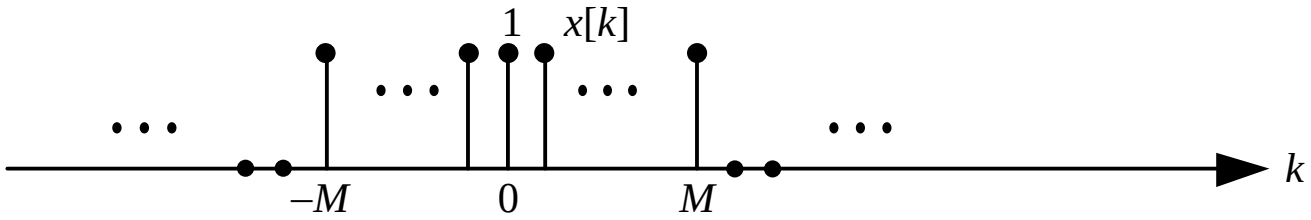


$$=\frac{\mathrm{e}^{\mathrm{j}\frac{2\pi}{6}2m}-\mathrm{e}^{-\mathrm{j}\frac{2\pi}{6}3m}}{1-\mathrm{e}^{-\mathrm{j}\frac{2\pi}{6}m}}=\frac{\sin\left(\frac{5\pi}{6}m\right)}{\sin\left(\frac{\pi}{6}m\right)}$$



$$m=0,1,\cdots,5$$

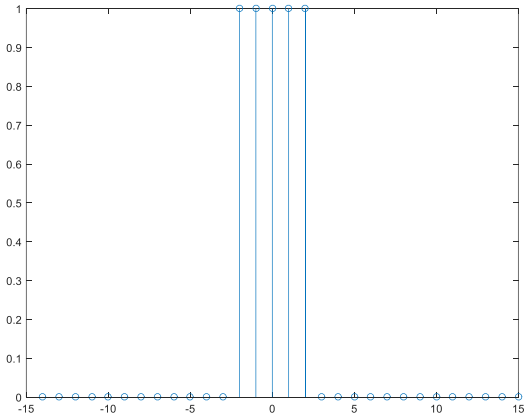
The frequency spectrum of a rectangular sequence with a period of N $(N>2M+1)$。

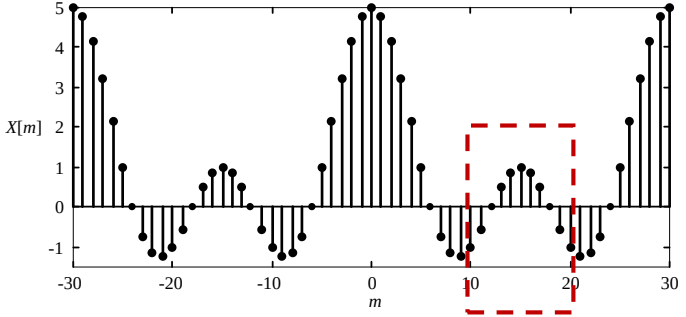


$$\tilde{X}[m]=\frac{\sin\left(\pi m(2M+1)/N\right)}{\sin\left(\pi m/N\right)}$$

DFS representation of discrete periodic signals

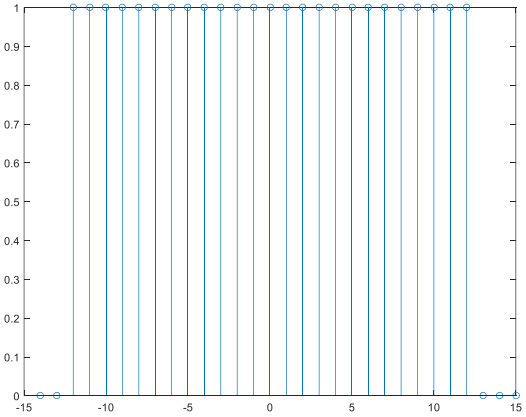
$N = 30,\ M = 2$

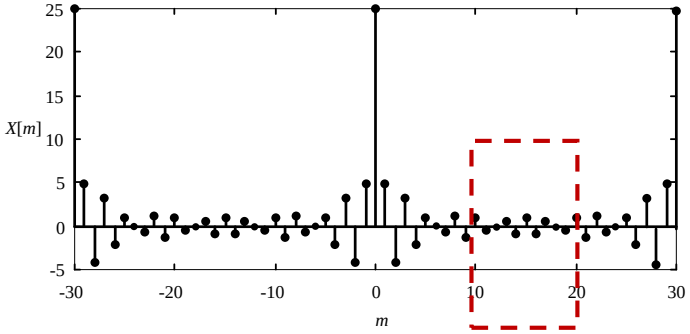




Periodic rectangular wave spectrum

$N = 30,\ M = 12$





Periodic rectangular wave spectrum

Properties of Discrete Periodic Signal DFS

※ Linear characteristics

※ Shift characteristics

※ Symmetrical characteristics

※ Periodic convolution characteristics

(1) Linear characteristics

If:

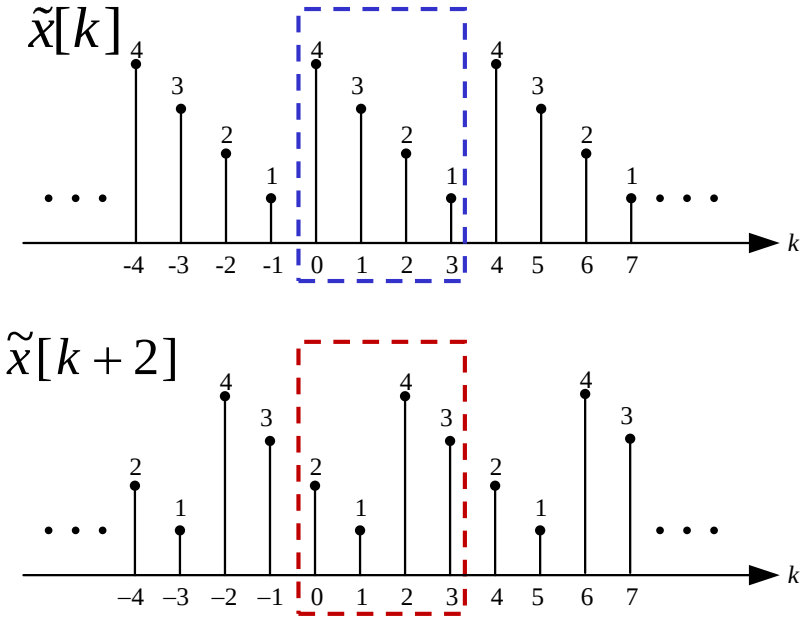
$$\tilde{x}_1[k] \xleftrightarrow{\text{DFS}} \tilde{X}_1[m]$$

$$\tilde{x}_2[k] \xleftrightarrow{\text{DFS}} \tilde{X}_2[m]$$

Then:

$$a\tilde{x}_1[k]+b\tilde{x}_2[k] \xleftrightarrow{\text{DFS}} a\tilde{X}_1[m]+b\tilde{X}_2[m]$$

(2) Shift characteristics



A periodic sequence remains to be a periodic sequence of the same period after being shifted, so it is only necessary to observe the sequence for one period after the shift.

(2) Shift characteristics

- Time domain displacement characteristics

$$\tilde{x}[k-n] \xleftrightarrow{\text{DFS}} \tilde{X}[m]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mn}$$

The displacement of a sequence in the time domain corresponds to its phase shift in the frequency domain.

(2) Shift characteristics

- Frequency domain displacement characteristics

$$\tilde{x}[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}lk} \xleftrightarrow{\text{DFS}} \tilde{X}[m+l]$$

The phase shift of a sequence in the time domain corresponds to its displacement in the frequency domain

(3) Symmetrical characteristics

If $\tilde{x}[k] \xleftrightarrow{\text{DFS}} \tilde{X}[m]$

Then $\tilde{x}^*[k] \xleftrightarrow{\text{DFS}} \tilde{X}^*[-m]$

$\tilde{x}^*[-k] \xleftrightarrow{\text{DFS}} \tilde{X}^*[m]$

If $\tilde{x}[k]$ is a real sequence, and $\tilde{x}[k] = \tilde{x}^*[k]$ then at this time

$$\tilde{X}[m] = \tilde{X}^*[-m]$$

(3) Symmetrical characteristics

If $\tilde{x}[k]$ is a real sequence, then there are

$$\tilde{X}[m] = \tilde{X}^*[-m]$$

Amplitude and phase

$$\tilde{X}[m] = \left|\tilde{X}[m]\right| \mathrm{e}^{\mathrm{j}\varphi[m]}$$

$$\left|\tilde{X}[m]\right| = \left|\tilde{X}[-m]\right|$$

$$\varphi[m] = -\varphi[-m]$$

Real and imaginary parts

$$\tilde{X}[m] = \tilde{X}_{\mathrm{R}}[m] + \mathrm{j}\,\tilde{X}_{\mathrm{I}}[m]$$

$$\tilde{X}_{\mathrm{R}}[m] = \tilde{X}_{\mathrm{R}}[-m]$$

$$\tilde{X}_{\mathrm{I}}[m] = -\tilde{X}_{\mathrm{I}}[-m]$$

(3) Symmetrical characteristics

If $\tilde{x}[k]$ is a real sequence, then there are $\tilde{X}[m]=\tilde{X}^{*}[-m]$

$$\tilde{X}[m]=\left|\tilde{X}[m]\right|\mathrm{e}^{\mathrm{j}\varphi[m]}$$

$$\tilde{X}[-m]=\left|\tilde{X}[-m]\right|\mathrm{e}^{\mathrm{j}\varphi[-m]}$$

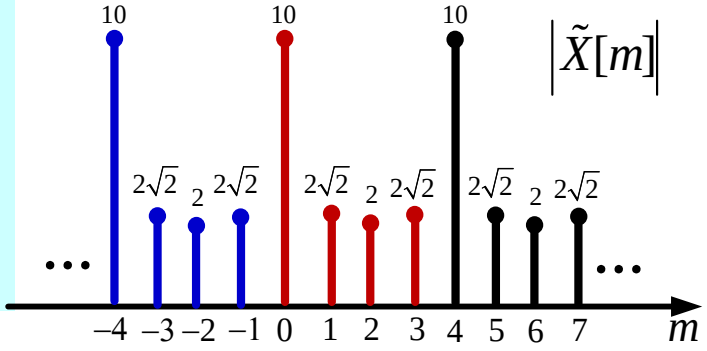
$$\tilde{X}^{*}[-m]=\left|\tilde{X}[-m]\right|\mathrm{e}^{-\mathrm{j}\varphi[-m]}$$

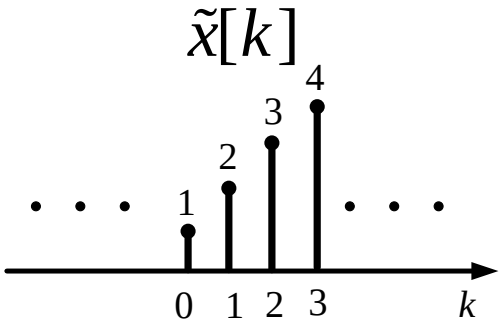
$$\left|\tilde{X}[m]\right|=\left|\tilde{X}[-m]\right|$$

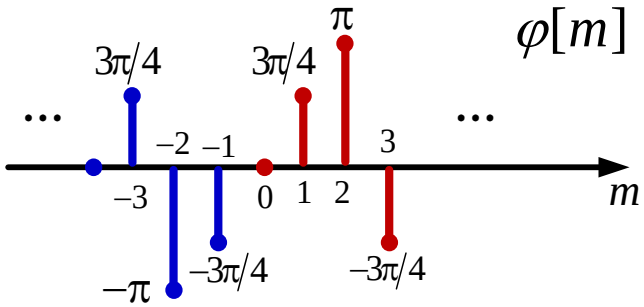
$$\varphi[m]=-\varphi[-m]$$

Properties of Discrete Periodic Signal DFS

If $\tilde{x}[k]$ is a real sequence, its amplitude spectrum is even symmetric and its phase spectrum is odd symmetric.



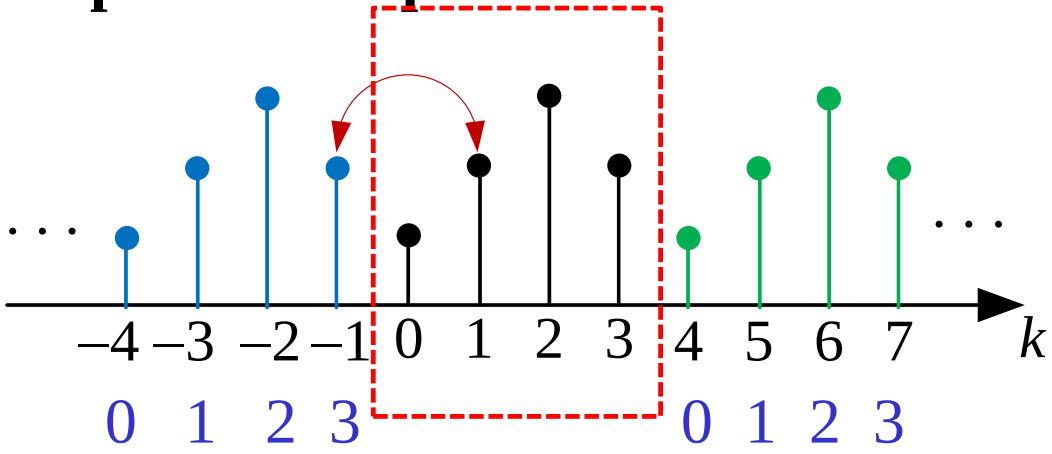


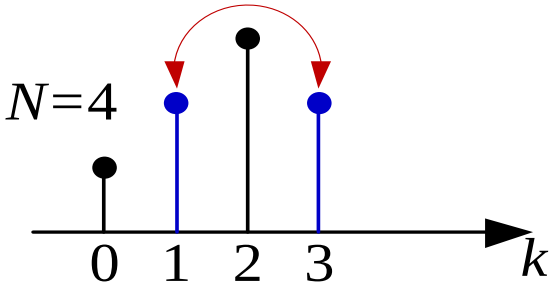


(3) Symmetrical characteristics

Even symmetry of periodic sequences:

$$\tilde{x}[k]=\tilde{x}[-k]=\tilde{x}[N-k]$$

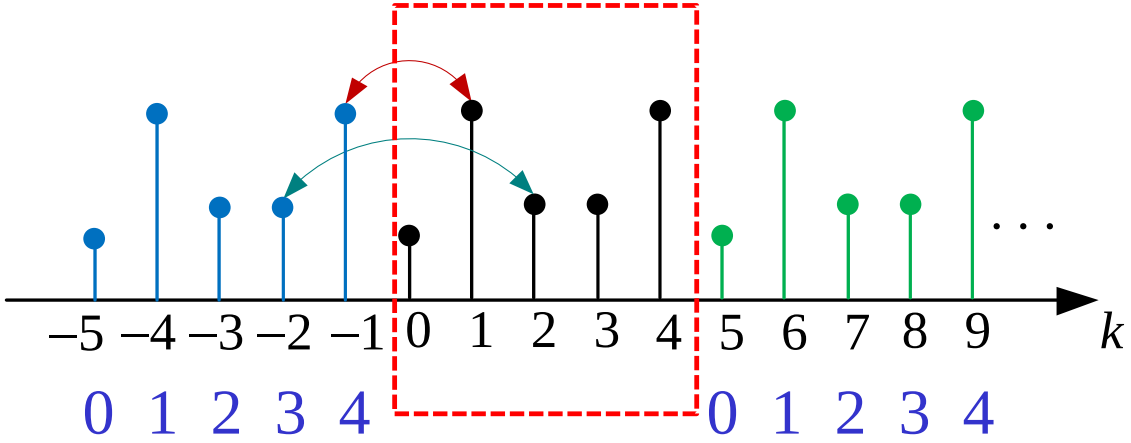


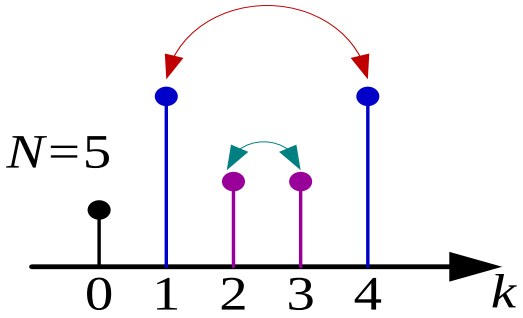


(3) Symmetrical characteristics

Even symmetry of periodic sequences:

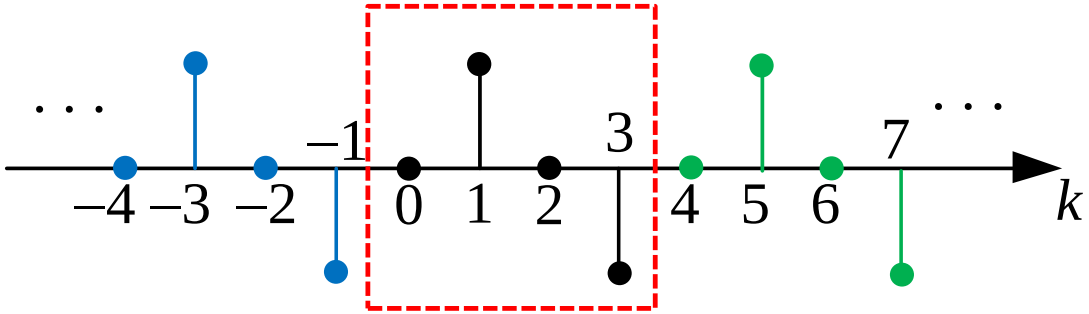
$$\tilde{x}[k]=\tilde{x}[-k]=\tilde{x}[N-k]$$

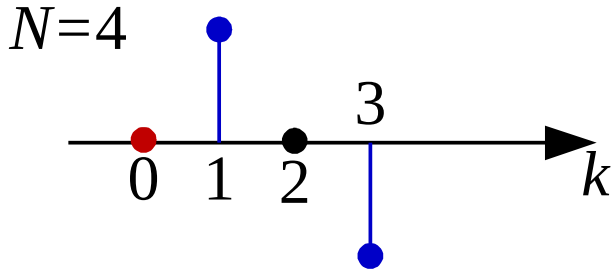




(3) Symmetrical characteristics

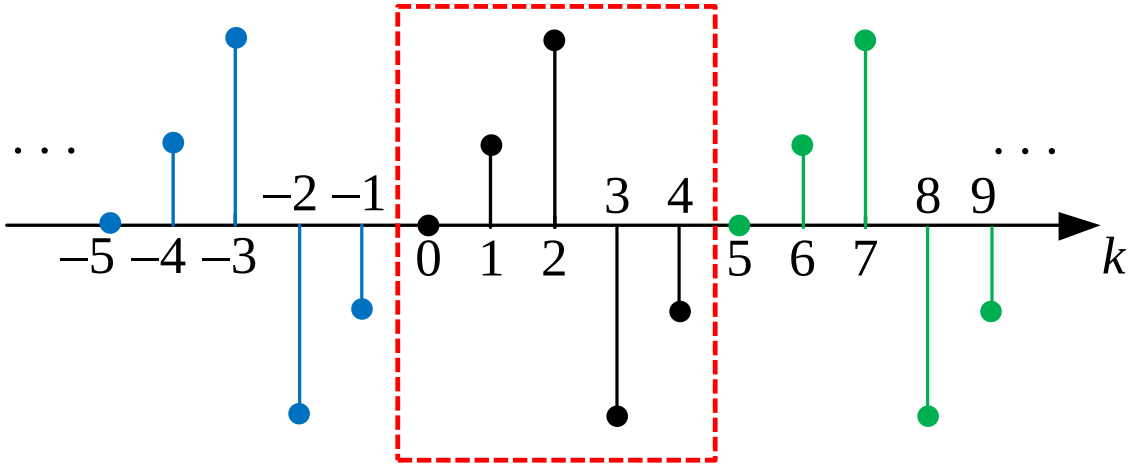
Odd symmetry of periodic sequences: $\tilde{x}[k]=-\tilde{x}[-k]=-\tilde{x}[N-k]$

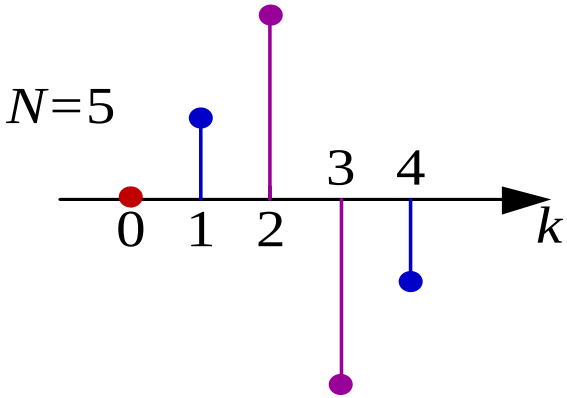




(3) Symmetrical characteristics

Odd symmetry of periodic sequences: $\tilde{x}[k] = -\tilde{x}[-k] = -\tilde{x}[N-k]$





(4) Definition of Periodic Convolution

$$\tilde{x}_1[k] \tilde{*} \tilde{x}_2[k] = \sum_{n=0}^{N-1} \tilde{x}_1[n]\tilde{x}_2[k-n]$$

※ Periodic convolution is the convolution operation of two equally periodic sequences.

※ The result of periodic convolution is still a periodic sequence with the same period.

The difference from linear convolution: only summing within one cycle

Properties of Discrete Periodic Signal DFS

(4) Periodic Convolution Theorem

※ Time domain periodic convolution theorem:

$$\tilde{x}_1[k] \tilde{*} \tilde{x}_2[k] \xleftarrow{DFS} \tilde{X}_1[m] \cdot \tilde{X}_2[m]$$

※ Frequency domain periodic convolution theorem:

$$\tilde{x}_1[k] \cdot \tilde{x}_2[k] \xleftarrow{DFS} \frac{1}{N} \tilde{X}_1[m] \tilde{*} \tilde{X}_2[m]$$

The periodic convolution in the time domain corresponds to the product of the frequency domain;

The product in the time domain corresponds to the periodic convolution in the frequency domain.

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.